External longitudinal validation of metabolic syndrome risk scores for cardiovascular, diabetes, and all-cause mortality in the US NHANES Linked Mortality File

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Abstract

Importance. Two metabolic syndrome risk scoring methods, a continuous score based on triangular areal similarity (RMRS) and a decision-tree formulation, have been developed and internally validated on Korean cohorts. External validation in a US cohort against established clinical risk equations, with long-term mortality outcomes, has not been reported.

Objective. To externally validate the RMRS and the metabolic syndrome decision tree against the ACC/AHA Pooled Cohort Equations (PCE), Framingham 2008, and FINDRISC for cardiovascular, diabetes-related, and all-cause mortality in a representative US adult population.

Design, setting, and participants. Prospective cohort using NHANES survey cycles 1999 through 2018 linked to the public-use NHANES Linked Mortality File (2019 release). Adults aged 20 to 79 years with non-missing data on the fasting metabolic syndrome components were eligible. Cohort N and exclusions reported in the Results.

Exposures. Five risk scores computed at NHANES baseline: RMRS, the metabolic syndrome decision tree, PCE, Framingham 2008, and FINDRISC.

Main outcomes and measures. Cardiovascular mortality, diabetes-related mortality (broadened definition; see Methods), and all-cause mortality, ascertained via the Linked Mortality File. Discrimination was quantified by inverse-probability-of-censoring weighted (IPCW) time-dependent AUC at 5 years and at off-cap horizons (9.5 years for cardiovascular and all-cause; 14.5 years for diabetes-related). Calibration, net benefit (decision curve analysis), pairwise delta-AUC, and incremental value (continuous NRI and IDI) were reported per registered analysis plan. Confidence intervals were computed via PSU-cluster bootstrap at 500 resamples.

Results. The pooled fasting-subsample cohort comprised 17,031 adults aged 20 to 79 years, with 806 all-cause deaths within the 10-year follow-up cap. Cardiovascular and diabetes-related (cause-specific) analyses used the 13,836 participants from cycles with full leading-cause coding (1999 to 2014), among whom 173 cardiovascular deaths accrued within the 10-year cap and 77 diabetes-related deaths under the broadened definition within the 15-year cap. For cardiovascular and all-cause mortality at the 9.5-year horizon, the clinical risk equations clearly dominated the metabolic-syndrome-derived scores; Framingham achieved a bootstrap-estimated AUC of 0.858 (95% CI 0.831 to 0.884) for cardiovascular mortality and 0.810 (0.795 to 0.825) for all-cause mortality, compared with RMRS at 0.660 (0.625 to 0.693) and 0.600 (0.578 to 0.622). For diabetes-related mortality at the 14.5-year horizon, RMRS and FINDRISC were comparable (RMRS 0.752 [0.696 to 0.804] vs FINDRISC 0.770 [0.705 to 0.826]), and both clearly exceeded the metabolic syndrome decision tree at 0.582 (0.532 to 0.631). The registered H3 non-inferiority test (RMRS vs FINDRISC for diabetes-related mortality, margin -0.05 AUC) yielded a point delta-AUC of $+0.014$ favoring RMRS with a bootstrap 95% CI of $(-0.051, +0.090)$, whose lower bound fell 0.001 AUC units below the registered margin; the DeLong-based CI was similar at $(-0.053, +0.080)$. Adding RMRS on top of FINDRISC for diabetes-related mortality produced an IDI of 0.0042 (0.0006 to 0.0128) and a continuous NRI of 0.139 (0.023 to 0.274). An XGBoost ceiling on the same five components placed an upper bound of 0.828 AUC for diabetes-related mortality at the 14.5-year horizon, roughly 6 to 8 AUC points above RMRS and FINDRISC.

Conclusions and relevance. Metabolic-syndrome-derived risk scores do not substitute for established clinical risk equations when the outcome is cardiovascular or all-cause mortality. For diabetes-related mortality, RMRS matches the dedicated FINDRISC screening tool on point discrimination, although the registered non-inferiority margin was not formally established by the bootstrap CI. The decision-tree formulation underperformed the continuous RMRS on every outcome, suggesting that the underperformance pattern reflects the tree structure rather than the metabolic syndrome signal.

Introduction

Metabolic syndrome is a clustering of central adiposity, dysglycemia, dyslipidemia, and elevated blood pressure that is associated with increased risk of cardiovascular disease, type 2 diabetes, and all-cause mortality [NCEP Expert Panel, 2002, Alberti et al., 2009]. Several scoring frameworks have been proposed to quantify metabolic syndrome severity on a continuous scale or via decision-tree classification, with the aim of improving prevention beyond the binary NCEP ATP III or IDF criteria.

Two such frameworks have recently been developed and internally validated on Korean cohorts. The Robust Metabolic Syndrome Risk Score (RMRS) maps the five component features into a continuous risk index via triangular areal similarity [Shin et al., 2024]. A complementary decision-tree formulation segments the input space using the same five component features [Shin et al., 2023]. The two papers together constitute the most-cited line of research in this group's metabolic syndrome work, with the decision-tree paper being the highest-cited.

External validation of these scores in a non-Asian population, against established clinical prediction equations, on long-term mortality outcomes, has not been reported. The ACC/AHA Pooled Cohort Equations [Goff et al., 2014] and the Framingham 2008 general cardiovascular risk profile [D'Agostino et al., 2008] are the primary baselines for cardiovascular risk prediction in US clinical practice. FINDRISC [Lindstrom and Tuomilehto, 2003] is the most widely used non-invasive screening tool for incident type 2 diabetes. A head-to-head longitudinal comparison on a representative US cohort would clarify the place, if any, of metabolic-syndrome-derived scores within the existing prediction-tool ecosystem.

We performed this comparison using NHANES survey cycles 1999 through 2018 linked to the public-use Linked Mortality File (2019 release) [National Center for Health Statistics, 2019]. The analysis was pre-registered on OSF, follows TRIPOD+AI [Collins et al., 2024] and STROBE [von Elm et al., 2007] reporting standards, and reports discrimination, calibration, net benefit, pairwise delta-AUC, and incremental value for three mortality outcomes (cardiovascular, diabetes-related, all-cause) across all five risk scores. The registered primary non-inferiority hypothesis (H3) tested whether RMRS is non-inferior to FINDRISC for predicting diabetes-related mortality at the 14.5-year horizon, with a registered margin of -0.05 AUC.

Methods

Study design and data sources

We used a prospective cohort design on the public-use NHANES survey [National Center for Health Statistics, 2019] cycles A through J (calendar years 1999 through 2018) linked to the National Center for Health Statistics public-use Linked Mortality File (LMF) 2019 release. Both files are de-identified, publicly available, and do not require IRB approval for secondary analysis. NHANES employs a complex multi-stage probability sample of the US civilian, non-institutionalized population with sampling stratification (`sdmvstra`), primary sampling units (`sdmvpsu`), and per-cycle survey weights. All analyses incorporated the survey design via the R `survey` package; effective survey weights were rescaled to the pooled cohort following NCHS guidance.

The LMF provides mortality follow-up time in months (PERMTH_EXM) and the NCHS 113-cause underlying cause-of-death recode (UCOD_LEADING), plus contributing-cause flags for diabetes and hypertension.

Participants and inclusion criteria

Eligible participants were adults aged 20 to 79 years at NHANES examination with non-missing values on the metabolic syndrome components (waist circumference, fasting glucose, HDL cholesterol, triglycerides, and blood pressure) and no prior cardiovascular disease or diagnosed diabetes. The 20 to 79 window matches the pre-registered eligibility criterion; PCE was computed only on its applicable 40 to 79 subset (reported separately). Because triglycerides and HDL are measured only in the morning fasting subsample, the cohort is restricted to fasting-subsample participants, which removes approximately half of adult NHANES respondents. The HDL column name varies across cycles (LBDHDL in cycles A and B, 1999 to 2002; LBXHDD in cycle C, 2003 to 2004; LBDHDD in cycles D onward, 2005 to 2018) and was coalesced. Race and ethnicity were coalesced from RIDRETH3 (available from cycle G, 2011 onward) and RIDRETH1 (earlier cycles), with the limitation that the non-Hispanic Asian category is not separately identifiable before cycle G.

Risk scores

Five risk scores were computed at NHANES baseline.

RMRS [Shin et al., 2024] maps the five metabolic syndrome components to a continuous index via triangular areal similarity to a reference healthy point. Implementation follows the published formula without modification.

Metabolic syndrome decision tree [Shin et al., 2023] classifies subjects into terminal nodes via a published CART. The published Figure 8(A) was transcribed; the left subtree (BPWC_add ≤ 0.66) is implemented as a single safety-zone leaf with the published probability of $p = 0.137$. A sensitivity analysis refitting the left subtree on a held-out NHANES split was planned (see Sensitivity analyses).

ACC/AHA Pooled Cohort Equations (PCE) [Goff et al., 2014] estimate 10-year atherosclerotic cardiovascular disease risk. The four sex- and race-specific equations were implemented per the published coefficients. PCE applies to adults aged 40 to 79 without prior atherosclerotic cardiovascular disease; subjects outside this range or with missing race were excluded from PCE-specific analyses (PCE-eligible

N reported in Results).

Framingham 2008 general cardiovascular risk profile [D'Agostino et al., 2008] estimates 10-year general cardiovascular risk. Coefficients from the original publication were applied.

FINDRISC [Lindstrom and Tuomilehto, 2003] is an integer screening score (range 0 to 26) for incident type 2 diabetes risk. NHANES does not collect FINDRISC items directly; the implementation maps NHANES variables onto FINDRISC items as follows. Age, body mass index, and waist circumference map directly. Physical activity is derived from the physical activity questionnaire (PAQ): a positive response to any moderate or vigorous recreational or work activity item marks the participant as active, using PAD200/PAD320 for cycles A through D (1999 to 2006) and PAQ650/PAQ665/PAQ605/PAQ620 for cycles E through J (2007 to 2018), which span the 2007 PAQ redesign. Antihypertensive medication use maps from BPQ050A. Personal history of high blood glucose is taken as a positive prediabetes report (DIQ160, available cycles D onward) or a measured impaired fasting glucose of 100 to 125 mg/dL, the latter available in every cycle. Family history of diabetes maps from MCQ300C (available cycles D onward), mapped to the first-degree tier when positive. The daily vegetable and fruit item has no cycle-consistent NHANES counterpart and is held at the no-credit level; its omission is the one residual proxy limitation and is noted in the Discussion. MCQ300C and DIQ160 are absent before cycle D (1999 to 2004), so family history defaults to none and the high-blood-glucose item relies on the measured fasting-glucose component in those cycles.

Outcomes

Three mortality outcomes were defined.

Cardiovascular mortality was defined as a leading underlying cause of disease of the heart (UCOD_LEADING code 1) or cerebrovascular disease (code 5). The follow-up cap was 10 years; AUC was reported at $t = 5$ and at the off-cap horizon $t = 9.5$ years (AUC at exactly the cap is degenerate under IPCW). The 2015-2016 and 2017-2018 public-use files release only leading-cause codes 1, 2, and 10, folding cerebrovascular and other specific causes into "all other"; cardiovascular and diabetes-related (cause-specific) analyses were therefore restricted to cycles A through H (1999 to 2014), which carry the full leading-cause coding. All-cause mortality used every cycle.

All-cause mortality was defined as any death recorded in the LMF. The follow-up cap and horizons matched the cardiovascular outcome, using the complete 1999 to 2018 cohort.

Diabetes-related mortality used a broadened definition: UCOD_LEADING for diabetes mellitus (code 7), OR UCOD_LEADING for nephritis (code 9), OR the LMF DIABETES contributing-cause flag equal to 1. The follow-up cap was extended to 15 years to accumulate adequate events under the broadened definition; AUC horizons were $t = 5$, $t = 10$, and $t = 14.5$. The pre-registered narrow definition (UCOD_LEADING = 7 only) yielded 6 events at 10 years and was underpowered for Fine-Gray modeling; the broadening to include nephritis and the contributing-cause flag, plus the follow-up extension, was registered as an amendment to the OSF protocol prior to running survival models on the diabetes outcome.

Statistical analysis

For cardiovascular and diabetes-related mortality, we modeled cumulative incidence under competing risks using survey-weighted Fine-Gray subdistribution hazards [Fine and Gray, 1999], fit via the Geskus weighted-risk-set construction (`survival::finegray`) with the NHANES examination weight carried into the subdistribution Cox model and Huber-White standard errors clustered on the primary sampling unit. Subdistribution hazard ratios are reported per 1 standard deviation of each score. For all-cause mortality, we used survey-weighted Cox proportional hazards. Time-dependent discrimination was quantified with IPCW AUC [Blanche et al., 2013] via the `timeROC` R package with `iid = FALSE` (the `iid = TRUE` setting allocates an $N \times N$ matrix that exceeds available memory at the realized cohort size; see Software note in the Supplement); the time-dependent AUC is the rank-based discrimination metric and is not survey-weighted.

Calibration was assessed via IPCW Brier scores and integrated Brier scores at each primary horizon (`riskRegression::Score`), alongside the predicted-risk distributions. Net benefit was reported via decision curve analysis [Vickers and Elkin, 2006] over a threshold range of 1% to 30%. For the competing-risks outcomes (cardiovascular, diabetes-related), each score was recalibrated to a horizon-specific absolute risk via the Fine-Gray cumulative incidence function, and net benefit was computed with the Aalen-Johansen cause-specific incidence within the treat-positive group so that censoring and competing deaths are handled rather than ignored; for all-cause mortality the recalibration used a Cox model and the Kaplan-Meier complement. Decision thresholds for the single-threshold bootstrap summaries were placed on each outcome's achievable predicted-risk range (5% for all-cause, 2% for cardiovascular, 1% for diabetes-related), since cardiovascular- and diabetes-mortality predicted risks do not reach the higher thresholds used for composite cardiovascular event scores.

For the registered pairwise comparisons (H1: RMRS vs B9 on each outcome; H2: PCE vs Framingham on cardiovascular; H3: RMRS vs FINDRISC on diabetes-related), delta-AUC point estimates were taken from the IPCW `timeROC` fit; the corresponding 95% CIs and DeLong p-values were computed via `pROC::roc.test` on the horizon-cap binary outcome (subjects censored before the horizon were excluded from the DeLong test). Definitive confidence intervals for all primary discrimination and net-benefit quantities were computed via PSU-cluster bootstrap at 500 resamples, sampling whole `sdmvstra` $\times$ `sdmvpsu` clusters with replacement (about 301 clusters across pooled cycles). The bootstrap distribution is the registered sensitivity for the H3 non-inferiority test against the -0.05 AUC margin.

Incremental value of adding RMRS on top of FINDRISC for diabetes-related mortality was reported as continuous NRI [Pencina et al., 2008] and IDI, computed via `survIDINRI::IDI.INF` with 200 perturbation iterations.

The analysis was complete-case on the metabolic syndrome components that define eligibility. Scores requiring additional inputs were set to missing where those inputs were unavailable (for example, PCE and Framingham require total cholesterol), and the affected score's analyses were run on its non-missing subset; the per-score analytic N is reported with each result.

Sensitivity analyses

The following sensitivity analyses were pre-registered (OSF section 8) and either completed or in progress.

ML upper-bound benchmark. An XGBoost classifier trained on the five metabolic syndrome components (waist circumference, systolic and diastolic blood pressure, HDL cholesterol, fasting glucose, triglycerides) plus age and sex, with PSU $\times$ stratum-grouped 5-fold cross-validation, was used as a non-parametric ceiling for the achievable AUC from the same inputs. Hyperparameters were not tuned beyond defaults.

B9 left-subtree refit. A CART was refit on a held-out NHANES split to test whether the original B9 left-subtree approximation contributes to the observed B9 underperformance. (Planned; status in `prereg/osf-preregistration-draft.md`.)

Subgroup analyses. AUC and Fine-Gray models were planned within strata defined by sex, race/ethnicity, and age band (40 to 49, 50 to 59, 60 to 69, 70 to 79 years). The non-Hispanic Asian category is reported only for cycles G onward. (Planned.)

Bootstrap definitive CIs. PSU-cluster bootstrap at 500 resamples for the focal AUCs and delta-AUCs is the registered definitive sensitivity for H3 and for all primary discrimination contrasts. (In progress at manuscript draft time.)

Reporting standards and pre-registration

The study followed TRIPOD+AI [Collins et al., 2024] and STROBE [von Elm et al., 2007] reporting checklists; completed checklists are provided in the Supplement. The analysis plan was pre-registered on the Open Science Framework before any survival model was fit on the data; the registered protocol is available at the OSF identifier given in the Data and Code Availability section.

Software

Analyses were conducted in R 4.3.3 on Ubuntu 24.04. Key packages included `survey` (NHANES design), `riskRegression` (Fine-Gray subdistribution model and cumulative incidence prediction), `survival` (weighted-risk-set Fine-Gray, Cox, Aalen-Johansen incidence), `timeROC` (IPCW AUC), `pROC` (DeLong delta-AUC), and `survIDINRI` (NRI and IDI). Decision curve net benefit was computed directly from the recalibrated risks and the Aalen-Johansen incidence. The XGBoost sensitivity used Python 3.11 with `xgboost` and `scikit-learn`. All code is available at the public repository listed in Data and Code Availability.

Results

Cohort

The pooled NHANES 1999 to 2018 fasting subsample after applying the inclusion criteria comprised 17,031 adults aged 20 to 79 years. Of these, 9,815 were within the PCE age applicability window (40 to 79 years) and contributed to PCE-specific analyses. Within the 10-year follow-up cap, 806 all-cause deaths accrued in the full cohort. The cause-specific cardiovascular and diabetes-related analyses used the 13,836 participants from the 1999 to 2014 cycles with full leading-cause coding (7,904 of them PCE-eligible); within this subcohort, 173 cardiovascular deaths accrued within the 10-year cap, and 77 diabetes-related deaths met the broadened definition (UCOD_LEADING for diabetes mellitus or nephritis, or the LMF DIABETES contributing-cause flag) within the 15-year cap. Survey-weighted baseline characteristics, overall and by all-cause mortality status, are shown in Table 1. Decedents were older (weighted mean 59.2 versus 43.1 years), more often current smokers (35.4% versus 20.9%), and more often on antihypertensive treatment (31.9% versus 14.9%) than survivors.

Table 1: Survey-weighted baseline characteristics of the pooled NHANES 1999–2018 fasting subsample (N = 17,031), overall and stratified by all-cause mortality status within the 10-year follow-up cap. Continuous variables are reported as survey-weighted mean (standard deviation); categorical variables as weighted percentage. Counts are unweighted.

Characteristic	Overall (n = 17,031)	Alive (n = 16,225)	Death (n = 806)
Age, years	43.7 (15.0)	43.1 (14.7)	59.2 (14.6)
Body mass index, kg/m^2	28.3 (6.4)	28.3 (6.4)	28.5 (6.8)
Waist circumference, cm	96.9 (15.9)	96.8 (15.8)	101.4 (17.0)
Systolic blood pressure, mmHg	119.8 (16.0)	119.5 (15.6)	129.6 (20.9)
Diastolic blood pressure, mmHg	71.1 (10.9)	71.1 (10.8)	70.9 (12.8)
Fasting glucose, mg/dL	99.1 (17.2)	98.9 (16.9)	105.4 (23.2)
HDL cholesterol, mg/dL	54.1 (16.2)	54.2 (15.9)	53.7 (21.6)
Triglycerides, mg/dL	128.0 (112.4)	127.3 (113.1)	144.3 (93.2)
Total cholesterol, mg/dL	197.2 (41.0)	197.1 (40.9)	201.2 (44.6)
Female sex	51.3	51.7	40.0
Non-Hispanic White	68.7	68.4	77.4
Non-Hispanic Black	10.2	10.1	10.7
Mexican American	8.6	8.8	3.7
Non-Hispanic Asian	2.3	2.4	0.6
Current smoker	21.4	20.9	35.4
Antihypertensive treatment	15.6	14.9	31.9
Physically active	72.8	73.5	55.7
Family history of diabetes	25.9	26.0	22.6
Prior high glucose / prediabetes	38.9	38.7	43.1

Discrimination

Time-dependent AUCs at the primary horizons are summarized in Table 2. For all-cause mortality at the 9.5-year horizon, discrimination ranged from 0.525 (B9 decision tree) to 0.810 (Framingham 2008). For cardiovascular mortality at the same horizon, the range was 0.560 (B9) to 0.858 (Framingham). For diabetes-related mortality at the 14.5-year horizon, the range was 0.582 (B9) to 0.770 (FINDRISC), with RMRS at 0.752 sitting close to FINDRISC and well above B9.

Table 2: Time-dependent AUC at primary horizons with 95% PSU-cluster bootstrap CIs (500 resamples). All-cause AUCs use the full cohort (N = 17,031; FINDRISC N = 16,996 after listwise deletion on FINDRISC inputs). Cardiovascular and diabetes-related AUCs use the 1999 to 2014 cause-coded subcohort (N = 13,836; FINDRISC diabetes N = 13,806). PCE AUCs are computed on the PCE-eligible subset within each cohort (all-cause N = 9,815; cardiovascular N = 7,904).

Score	Outcome	Horizon	AUC (95% CI)
Framingham 2008	All-cause	9.5y	0.810 (0.795, 0.825)
PCE	All-cause	9.5y	0.758 (0.736, 0.778)
FINDRISC	All-cause	9.5y	0.672 (0.651, 0.695)
RMRS	All-cause	9.5y	0.600 (0.578, 0.622)
B9 tree	All-cause	9.5y	0.525 (0.505, 0.542)
Framingham 2008	Cardiovascular	9.5y	0.858 (0.831, 0.884)
PCE	Cardiovascular	9.5y	0.810 (0.772, 0.841)
RMRS	Cardiovascular	9.5y	0.660 (0.625, 0.693)
B9 tree	Cardiovascular	9.5y	0.560 (0.519, 0.593)
FINDRISC	Diabetes-related	14.5y	0.770 (0.705, 0.826)
RMRS	Diabetes-related	14.5y	0.752 (0.696, 0.804)
B9 tree	Diabetes-related	14.5y	0.582 (0.532, 0.631)

Calibration

IPCW Brier and integrated Brier scores at each primary horizon are provided in the Supplement. The metabolic-syndrome-derived scores are miscalibrated when read as raw probabilities (RMRS is bounded in a narrow band and FINDRISC is an integer count rather than a probability), which motivated the per-score recalibration to an absolute-risk scale used for the decision curve analysis below. Figure 1 shows predicted-versus-observed calibration on this recalibrated scale, with subjects binned by predicted risk and observed cumulative incidence estimated by Aalen-Johansen (cardiovascular and diabetes-related) or one minus Kaplan-Meier (all-cause). After recalibration the continuous scores track the diagonal across the risk range; the B9 decision tree maps subjects onto only a small number of distinct predicted-risk levels, so its calibration is summarized by two points.

Net benefit

Decision curve analysis figures for the three outcomes are provided as `results/dca_allcause.png`, `results/dca_cv.png`, and `results/dca_dm.png`. For all-cause mortality at the 9.5-year horizon and a 5% decision threshold, PCE produced the largest net benefit (0.0537 per person, 95% CI 0.0485 to 0.0589), followed by Framingham (0.0304, 0.0263 to 0.0345), FINDRISC (0.0176, 0.0138 to 0.0226), RMRS (0.0124, 0.0078 to 0.0175), and the B9 tree (0.0084, 0.0040 to 0.0136). For cardio-

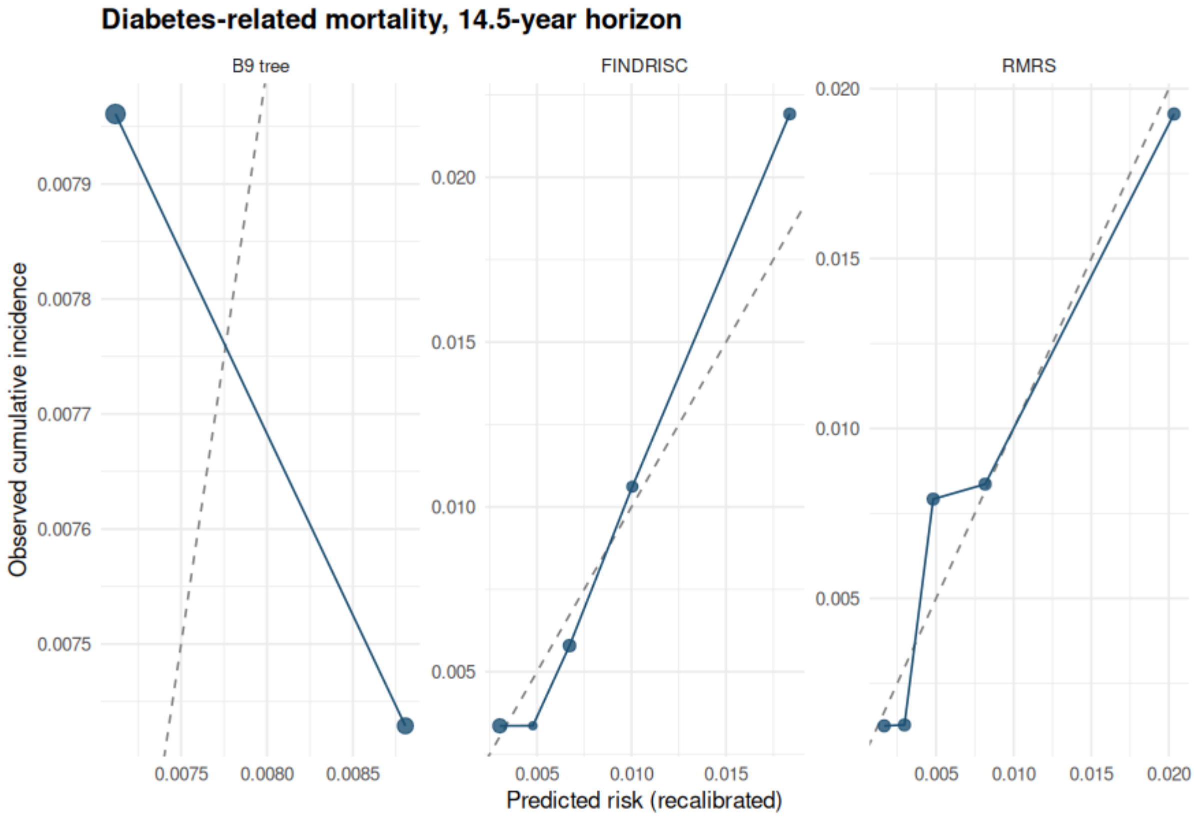

Figure 1: Calibration of recalibrated diabetes-related mortality risk at 14.5 years for RMRS, the B9 tree, and FINDRISC. Points are risk-bin means sized by bin count; the dashed line is perfect calibration. Calibration plots for all-cause and cardiovascular mortality are provided in the Supplement (`results/plots/calibration_allcause.png`, `calibration_cv.png`).

vascular mortality, predicted mortality risk does not reach the ACC/AHA 7.5% composite-event statin threshold, so net benefit was evaluated at 2%; PCE (0.0099, 0.0070 to 0.0129) and Framingham (0.0049, 0.0031 to 0.0074) led the metabolic-syndrome-derived scores, consistent with the AUC ordering. For diabetes-related mortality at a 1% threshold, FINDRISC (0.0023, 0.0002 to 0.0043) and RMRS (0.0017, 0.0006 to 0.0035) exceeded treat-none while the B9 tree (0.0000) did not.

Pairwise comparisons and incremental value

The five registered pairwise delta-AUC comparisons are summarized in Table 3.

For the H3 non-inferiority test of RMRS versus FINDRISC on diabetes-related mortality at the 14.5-year horizon, the bootstrap 95% percentile lower bound was -0.051, which fell below the registered margin of -0.05 by 0.001 AUC units. The point estimate of $+0.014$ favored RMRS. The DeLong-based 95% CI on the horizon-cap binary outcome was $(-0.053, +0.080)$, close to the bootstrap interval. By the registered decision rule, non-inferiority was not formally established, though the lower bound missed

Table 3: Registered pairwise delta-AUC comparisons. CIs are 95% PSU-cluster bootstrap percentile intervals (500 resamples); DeLong-based CIs on the horizon-cap binary outcome agreed to within about 0.01 AUC across the contrasts.

Contrast	Outcome (horizon)	Delta-AUC (95% CI)
RMRS vs FINDRISC (H3, non-inferiority)	Diabetes-related (14.5y)	$+0.014$ (-0.051, $+0.090$)
RMRS vs B9	All-cause (9.5y)	$+0.071$ ($+0.049$, $+0.094$)
RMRS vs B9	Cardiovascular (9.5y)	$+0.093$ ($+0.050$, $+0.135$)
RMRS vs B9	Diabetes-related (14.5y)	$+0.158$ ($+0.088$, $+0.220$)
PCE vs Framingham (H2)	Cardiovascular (9.5y)	$+0.007$ (-0.010, $+0.022$)

the margin by a margin within bootstrap resampling noise.

Adding RMRS on top of FINDRISC for diabetes-related mortality produced an IDI of 0.0042 (95% CI 0.0006 to 0.0128) and a continuous NRI of 0.139 (95% CI 0.023 to 0.274). Both intervals excluded zero, indicating that RMRS captured diabetes-mortality information beyond what FINDRISC already provided, with modest magnitude.

For the H2 comparison of PCE versus Framingham on cardiovascular mortality, the delta-AUC was $+0.007$ (95% CI -0.010 to $+0.022$) favoring PCE. The interval contained zero, indicating no detectable difference between the two clinical equations on this outcome in this cohort.

Sensitivity analyses

The XGBoost ceiling on the five metabolic syndrome components plus age and sex, with PSU $\times$ stratum-grouped 5-fold cross-validation, yielded AUCs of 0.813 for all-cause mortality at 9.5 years, 0.817 for cardiovascular mortality at 9.5 years, and 0.828 for diabetes-related mortality at 14.5 years. For all-cause mortality, the ML ceiling coincided with the Framingham AUC, indicating no headroom for a tree-based metabolic syndrome variant on this outcome. For cardiovascular mortality, the clinical Framingham equation at 0.858 exceeded the ML ceiling at 0.817, indicating that the metabolic syndrome inputs alone cannot reach the Framingham performance level even under flexible non-parametric modeling. For diabetes-related mortality, the ML ceiling at 0.828 exceeded RMRS at 0.752 and FINDRISC at 0.770, leaving roughly 6 to 8 AUC points of headroom for a tree-based variant of the metabolic syndrome representation specifically for diabetes mortality.

The transported B9 tree collapses its left subtree (joint BP and waist elevation below about 0.66) into a single baseline-risk leaf, because the published Figure 8(A) split values are not legible. To test whether this approximation, rather than the tree method itself, drives B9's underperformance, we refit a CART on a held-out 30% NHANES split using the same three BPWC features to predict ATP III metabolic syndrome, then scored the held-out subjects (Table 4). The data-driven tree independently routed the entire low-BPWC region to a single low-risk leaf (4.7% metabolic syndrome prevalence), so the transported tree's flat left subtree was not the source of its underperformance. The refit instead gained discrimination by re-estimating the right-subtree splits and leaf probabilities on US data: held-out AUC rose from 0.535 to 0.595 for all-cause, 0.561 to 0.647 for cardiovascular, and 0.606 to 0.783 for diabetes-related mortality, the last exceeding RMRS at 0.735 on the same held-out subjects. B9's gap relative to RMRS in the main analysis is therefore largely a transportability artifact of the Korean-calibrated splits, not an intrinsic limit of a tree on these features.

Table 4: B9 left-subtree US-refit sensitivity. Time-dependent AUC on a held-out 30% NHANES split for the transported B9 tree, a CART refit on the training split using the same three BPWC features to predict ATP III metabolic syndrome, and the continuous RMRS for reference. The data-driven tree independently collapses the left region (BPWC_add below about 0.69) into a single low-risk leaf, matching the transported tree's approximation there; the gain comes from refitting the right subtree to US data.

Outcome	Horizon	B9 transported	B9 US-refit	RMRS
All-cause	9.5y	0.535	0.595	0.582
Cardiovascular	9.5y	0.561	0.647	0.660
Diabetes-related	14.5y	0.606	0.783	0.735

The pre-registered subgroup analyses by sex, race/ethnicity, and age band for the diabetes-related H3 contrast are shown in Table 5. RMRS and FINDRISC tracked each other within every estimable stratum: their AUC difference stayed within about 0.05 in most strata, with RMRS ahead in the 50 to 59 and 60 to 69 age bands and FINDRISC ahead among women and in the oldest band. No subgroup reversed the overall near-equivalence. Strata with fewer than 10 diabetes-related deaths, including the non-Hispanic Asian group (present only from the 2011 cycle onward) and the youngest age bands, were not estimable and are omitted. For cardiovascular mortality, Framingham retained the highest AUC in every estimable sex, race/ethnicity, and age stratum, consistent with the pooled result; the full grid across all three outcomes is in `results/subgroup_auc.csv`.

Discussion

In this external longitudinal validation of two metabolic syndrome risk scoring methods (RMRS and the B9 decision tree) against three established clinical risk equations (PCE, Framingham 2008, FINDRISC) on the US NHANES Linked Mortality File, the headline finding is outcome-specific. The clinical risk

Table 5: Pre-registered subgroup time-dependent AUC for the diabetes-related mortality H3 contrast (RMRS versus FINDRISC) at 14.5 years, within strata of sex, race/ethnicity, and age band (1999 to 2014 cause-coded subcohort). Strata with fewer than 10 diabetes-related deaths are omitted because IPCW AUC is unstable below that count; the omitted strata include non-Hispanic Asian (available only from the 2011 cycle onward) and the youngest age bands. The full grid across all outcomes is in `results/subgroup_auc.csv`.

Stratum	Level	Events	RMRS AUC	FINDRISC AUC
Sex	Male	49	0.728	0.734
	Female	28	0.778	0.832
Race/ethnicity	Mexican American	21	0.707	0.744
	Non-Hispanic White	23	0.780	0.781
	Non-Hispanic Black	21	0.776	0.801
Age band	50-59	11	0.779	0.627
	60-69	26	0.676	0.622
	70-79	27	0.603	0.641

equations dominated the metabolic-syndrome-derived scores when the outcome was cardiovascular or all-cause mortality. For diabetes-related mortality, the metabolic-syndrome-derived RMRS competed with FINDRISC on point discrimination and substantially outperformed the B9 decision tree.

For diabetes-related mortality at the 14.5-year horizon, RMRS achieved an AUC of 0.752 against FINDRISC at 0.770, with a delta-AUC of $+0.014$ favoring RMRS and a bootstrap 95% percentile CI of $(-0.051, +0.090)$. The registered non-inferiority test at the -0.05 margin was not formally established: the bootstrap lower bound fell 0.001 AUC units below the margin, a gap smaller than the resampling resolution, and the DeLong interval gave the same qualitative result. Both intervals locate the two scores within a narrow band on this outcome. The IDI and continuous NRI for adding RMRS on top of FINDRISC excluded zero, indicating that RMRS captures diabetes-mortality information beyond the FINDRISC items, with a continuous NRI of 0.14. RMRS occupies a complementary niche in diabetes-mortality prediction alongside FINDRISC, which remains the primary screening tool of choice. This comparison now uses a FINDRISC implementation with NHANES-derived family history, prediabetes/impaired-fasting-glucose, and physical-activity items rather than constant placeholders, so the near-equivalence is not an artifact of a degraded comparator.

For cardiovascular mortality at the 9.5-year horizon, Framingham 2008 achieved an AUC of 0.858 and PCE achieved 0.810, with a delta-AUC of $+0.007$ between them (95% CI -0.010 to $+0.022$). The metabolic-syndrome-derived RMRS at 0.660 sat about 20 AUC points below these clinical equations, and the decision curve analysis showed the same ordering. For all-cause mortality, Framingham again led at 0.810, with PCE at 0.758 and the metabolic-syndrome-derived scores in the 0.525 to 0.600 range.

The XGBoost ceiling on the metabolic syndrome inputs alone, at 0.813 for all-cause and 0.817 for cardiovascular, did not close the gap. The metabolic syndrome features do not contain the cardiovascular-mortality and all-cause-mortality information that the clinical equations integrate, regardless of the modeling approach used on them.

The B9 decision tree formulation consistently underperformed the continuous RMRS on every outcome, with delta-AUCs against RMRS of $+0.071$ for all-cause, $+0.093$ for cardiovascular, and $+0.158$ for diabetes-related mortality (all CIs excluded zero). The XGBoost ceiling, fit on the same five metabolic syndrome inputs, exceeded both the continuous RMRS and the B9 tree on diabetes mortality (0.828 vs 0.752 vs 0.582), showing the metabolic syndrome signal itself is not the binding constraint. The pre-registered US-refit sensitivity (Table 4) localizes the cause of the B9 gap: a CART refit on a held-out NHANES split, using the same BPWC features, recovered diabetes-mortality discrimination of 0.783, above RMRS on the same held-out subjects, and the data-driven tree independently routed the low-BPWC region to a single low-risk leaf just as the transported tree does. The B9 underperformance is therefore largely a transportability artifact of the Korean-calibrated right-subtree splits and leaf probabilities, rather than a limitation of the tree method or of the collapsed left subtree.

Several limitations apply. First, the FINDRISC implementation maps most items to NHANES variables (physical activity from PAQ, family history from MCQ300C, prior high glucose from DIQ160 or measured impaired fasting glucose), but two gaps remain: the daily vegetable and fruit item has no cycle-consistent NHANES counterpart and is held at the no-credit level, and the family-history and self-reported-prediabetes items are only available from cycle D (2005 onward), so earlier cycles default those items. The residual proxy imprecision biases FINDRISC conservatively. Second, the cause-specific cardiovascular and diabetes-related analyses are restricted to the 1999 to 2014 cycles, because the public-use Linked Mortality File releases only three leading-cause categories for the 2015 to 2018 cycles and folds stroke and diabetes-as-underlying-cause into “all other”; all-cause mortality uses the full cohort. The excluded cycles contribute little cause-specific follow-up given their short observation window. Third, the cohort is restricted to the morning fasting subsample of NHANES, which removes approximately half of adult NHANES respondents and may introduce selection on factors associated with willingness or ability to fast. Fourth, the non-Hispanic Asian race category is only separately identifiable in NHANES cycles G onward (2011 to 2018), which limits subgroup precision for that group. Fifth, the diabetes-related mortality outcome uses a broadened definition (UCOD for diabetes mellitus or nephritis, or the LMF diabetes contributing-cause flag); the pre-registered narrow definition was underpowered for Fine-Gray modeling, and the broadening was registered as an amendment prior to running survival models on this outcome. The clinical implication is that the metabolic syndrome continuous risk score occupies a complementary niche in diabetes-mortality prediction (alongside FINDRISC), and does not substitute for established clinical risk equations for cardiovascular or all-cause mortality prediction. Future work should evaluate the metabolic syndrome inputs combined with the clinical risk equation inputs in joint models, and validate the diabetes-mortality finding in non-US cohorts.

Data and code availability

All analysis code is available at `https://github.com/jayhemnani9910/longitudinal-mets-validat` (MIT license). The OSF pre-registration is at (TBD: OSF identifier to be inserted on submission).

Conflicts of interest

The author reports no conflicts of interest.

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References

K. G. M. M. Alberti, Robert H. Eckel, Scott M. Grundy, et al. Harmonizing the metabolic syndrome. *Circulation*, 2009.

Paul Blanche, Jean-Francois Dartigues, and Helene Jacqmin-Gadda. Estimating and comparing time-dependent areas under receiver operating characteristic curves for censored event times with competing risks. *Statistics in Medicine*, 2013.

Gary S. Collins, Karel G. M. Moons, Paula Dhiman, et al. Tripod+ai statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 2024.

Ralph B. D'Agostino, Ramachandran S. Vasan, Michael J. Pencina, et al. General cardiovascular risk profile for use in primary care. *Circulation*, 2008.

Jason P. Fine and Robert J. Gray. A proportional hazards model for the subdistribution of a competing risk. *Journal of the American Statistical Association*, 1999.

David C. Goff, Donald M. Lloyd-Jones, Glen Bennett, et al. 2013 acc/aha guideline on the assessment of cardiovascular risk. *Circulation*, 2014.

Jaana Lindstrom and Jaakko Tuomilehto. The diabetes risk score: a practical tool to predict type 2 diabetes risk. *Diabetes Care*, 2003.

National Center for Health Statistics. Nhanes linked mortality file, public-use 2019 release. `https://www.cdc.gov/nchs/data-linkage/mortality-public.htm`, 2019.

NCEP Expert Panel. Third report of the national cholesterol education program expert panel on detection, evaluation, and treatment of high blood cholesterol in adults (adult treatment panel iii). *Circulation*, 2002.

Michael J. Pencina, Ralph B. D'Agostino, Ralph B. Jr. D'Agostino, and Ramachandran S. Vasan. Evaluating the added predictive ability of a new marker: from area under the roc curve to reclassification and beyond. *Statistics in Medicine*, 2008.

Hyo Jung Shin, Jaeyong Oh, and Simon S. Shim. Machine learning predictive model for metabolic syndrome prevention. *PLoS ONE*, 2023. B9 in the project corpus. Most-cited paper in Shim corpus.

Hyo Jung Shin, Jaeyong Oh, and Simon S. Shim. A robust metabolic syndrome risk score using triangular areal similarity. *PeerJ Computer Science*, 2024. B8 in the project corpus.

Andrew J. Vickers and Elena B. Elkin. Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making*, 2006.

Erik von Elm, Douglas G. Altman, Matthias Egger, et al. The strengthening the reporting of observational studies in epidemiology (strobe) statement. *Epidemiology*, 2007.